

Munich Delivery Operations

Analysis Summary: Meituan On-Demand Food Delivery Case Study

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Data Foundation

The dataset covers Munich delivery operations from January 2022 to December 2025, providing 1,461 daily records across 17 columns. Before any analysis could begin, the data required cleaning. Three rows where orders_delivered showed 88,888 were identified as dummy placeholders. On those same days, potential demand was only around 5,500, confirming the value was unrecoverable and the rows were removed. Three rows where total_time_mins showed 9,999 were sentinel values; these were rebuilt by summing assign_time_mins and order_handling_time_mins, which the dataset's own formula confirms. Three rows with a negative order_distance.km of -9.90 were reconstructed from the two sub-distance columns. Two rows where daily_online_restaurants were zero were forward-filled from the prior day; active restaurants on those same days numbered 542 and 839, confirming a recording error rather than an operational closure. Three missing delivery_rating values were filled using linear interpolation between neighbouring days. After cleaning, 1,458 rows remained with zero missing values across all columns.

In addition to the original 17 columns, 16 derived features were created including fulfillment_rate, lost_orders, orders_per_working_hour, sla_met_25, sla_met_30, is_weekend, day_of_week, and handling/assign time percentage splits. These were used throughout all five business questions.

Q1 · SLA and Service Quality: How Often Is the Target Met?

The dataset includes a delivery_time_target of 30 minutes, which represents the official SLA. Against this benchmark, Munich performs at 99.9%, with only one day in 1,458 exceeded 30 minutes. However, the more operationally meaningful target of 25 minutes reveals a very different picture. Only 838 of 1,458 days met the 25-minute threshold, giving a compliance rate of 57.5%. On the remaining 620 days the target was missed. The mean total delivery time across the clean dataset is 24.71 minutes, just 0.29 minutes below the 25-minute line, meaning the operation is running right at the edge. Any increase in demand or handling delay pushes Munich over.

Service quality was measured using a threshold of delivery_rating ≥ 3.5 . This value was chosen as a standard operations benchmark, selected based on the observed data range of 2.90 to 3.90, with 3.5 sitting at the midpoint as a reasonable quality floor. On this basis, the quality threshold was met on 74.2% of days. When both conditions are applied together, 25-minute delivery and rating at or above 3.5, only 57.0% of days satisfied both simultaneously.

Year-on-year 25-minute compliance

Year	Avg Delivery Time	25-Min Compliance	30-Min Compliance	Avg Rating
2022	24.08 mins	68.4%	100.0%	3.58
2023	24.43 mins	63.0%	100.0%	3.56
2024	25.04 mins	51.8%	99.7%	3.51
2025	25.27 mins	46.7%	100.0%	3.50

Compliance has declined every single year without recovery. Between 2022 and 2025 the rate fell from 68.4% to 46.7%, a drop of 21.7 percentage points. At the same time the average delivery time crossed the 25-minute mark in 2024 (25.04 mins) and worsened further in 2025 (25.27 mins), confirming the trend is structural rather than a one-off.

Root cause analysis

Delivery time splits into two components: assign_time_mins and order_handling_time_mins. Across the clean dataset, handling time accounts for approximately 78% of total delivery time (~19.3 mins) and assign time for the remaining 22% (~5.4 mins). Comparing days where the 25-minute target was met versus days where it was missed reveals the following gap:

Component	SLA Met Days	SLA Missed Days	Gap
Order Handling Time	18.34 mins	20.53 mins	+2.19 mins
Courier Assign Time	5.17 mins	5.79 mins	+0.62 mins

Handling time is the primary operational lever. A simulation confirms this: if handling time were reduced by 2 minutes on every day in the dataset, compliance would rise from 57.5% to **90.7%**, adding 484 days that would have met the target. No other single variable produces a comparable improvement. Among other operational factors correlated with total delivery time, order volume (+0.351) and working hours (+0.350) show the strongest association, followed by active restaurants (+0.304) and order distance (+0.137). Weekend status shows near-zero correlation (+0.035), meaning weekdays and weekends perform almost identically for SLA compliance.

Q2 · Impact of Working Hours on Order Fulfillment

The central question here is whether deploying more courier hours results in a higher proportion of potential orders being fulfilled. The answer, clearly supported by the data, is no. The Pearson correlation between daily working_hours and fulfillment_rate is +0.027, and the corresponding R-squared is 0.0008. Working hours explain less than 0.1% of fulfillment variation. To make this concrete, the dataset was split into five equal-sized bands from Very Low to Very High working hours:

Working-Hour Band	Avg Daily Hours	Avg Fulfillment Rate	Avg Efficiency (ord/hr)
Very Low (< 20%)	1,661 hrs	85.0%	2.18
Low (20 – 40%)	1,950 hrs	85.3%	2.15
Medium (40 – 60%)	2,213 hrs	85.2%	2.16
High (60 – 80%)	2,471 hrs	85.3%	2.13
Very High (> 80%)	2,899 hrs	85.2%	2.12

Despite a 75% increase in courier hours from the lowest to the highest band, fulfillment moves by only 0.30 percentage points across the entire range. The real driver of fulfillment is potential demand, which has a negative correlation of -0.153 with fulfillment rate. On days when market demand is unusually high, the system is overwhelmed regardless of how many couriers are working. City efficiency averaged 2.147 across the full four-year period and showed a monthly standard deviation of just ± 0.02 across all 48 months, confirming that adding couriers has not improved per-courier productivity either. Meanwhile, the absolute number of lost orders grew from 250,796 in 2022 to 354,660 in 2025, an increase of 41%, because market demand grew while the fulfillment rate stayed structurally fixed at around 85%.

Q3 · Order Volume Forecast: 120 Days (Jan–Apr 2026)

Before fitting any models, the time series properties of daily orders_delivered were examined. An Augmented Dickey-Fuller test returned a p-value of 0.4422, confirming the series is non-stationary and requires first-order differencing. Seasonal decomposition with a weekly period revealed a trend strength of 0.929, indicating very strong upward growth from approximately 3,719 to 5,999 orders per day over four years, and a seasonal strength of 0.781, confirming a clear and consistent weekly cycle. Four models were trained on data from January 2022 through September 2025 and evaluated on an identical 120-day held-out test set covering September 3 to December 31 2025.

Model	MAPE	Assessment
Holt-Winters ETS	4.70%	Selected: best accuracy on held-out test set
Ensemble (HW+Prophet+SARIMA)	5.86%	Average of three models, reduces individual bias
Prophet	5.95%	Good, slight overfitting on trend
SARIMA(1,1,1)(1,1,1,7)	6.51%	Good, struggles with non-linear trend growth

Holt-Winters ETS was selected as the planning model with a MAPE of 4.70%, which is considered excellent for a daily operational forecast. The model applies a damped trend component to avoid over-extrapolation. The 120-day forecast runs from January 1 to April 30 2026. Confidence Intervals were constructed as ± 1.96 times the residual standard deviation, giving a margin of ± 561 orders around the base forecast.

Month	Avg Orders/Day	Total Orders
Jan 2026	5,528	171,381
Feb 2026	5,501	154,029
Mar 2026	5,487	170,112
Apr 2026	5,475	164,238

Pessimistic: 4,937 orders/day (base minus CI margin);

Base: 5,498 orders/day (Holt-Winters mean, recommended for planning);

Optimistic: 6,059 orders/day (base plus CI margin).

Day-of-week average forecast (Jan–Apr 2026, Holt-Winters)

Day of Week	Avg Forecast (orders/day)	Day Type
Monday	5,034	Weekday
Tuesday	5,021	Weekday
Wednesday	5,072	Weekday
Thursday	5,137	Weekday
Friday	6,123	Weekday (peak)
Saturday	6,092	Weekend
Sunday	6,028	Weekend

Q4 · Driver Planning: 90 Days (Jan–Mar 2026)

Required drivers per day were estimated using a straightforward formula derived from city efficiency: Required Drivers = Forecasted Orders ÷ (City Efficiency × Opening Hours). City efficiency of 2.147 orders per working hour was taken directly from the data, the mean across all 1,458 clean days, with a monthly standard deviation of +/-0.02 confirming it is stable enough to use as a fixed planning constant. Opening hours differ by day type: weekdays average 11.90 hours and weekends 13.50 hours, both calculated from the daily_opening_hours column in the dataset.

Scenario	Avg Drivers/Day	Basis
Pessimistic	186	Pessimistic order forecast (4,937/day)
Base (recommended)	208	Holt-Winters base forecast (5,498/day)
Buffered (+10%)	228	Base + 10% contingency above MAPE
Optimistic	229	Optimistic order forecast (6,059/day)

Day-of-week driver requirements (base scenario, Jan–Mar 2026)

Day of Week	Avg Orders/Day	Opening Hrs	Base Drivers	Buffered (+10%)
Monday	5,034	11.90 hrs	197	217
Tuesday	5,021	11.90 hrs	197	217
Wednesday	5,072	11.90 hrs	199	219
Thursday	5,137	11.90 hrs	201	221
Friday	6,123	11.90 hrs	240	264
Saturday	6,092	13.50 hrs	210	231
Sunday	6,028	13.50 hrs	208	229

An important operational insight emerges from the day-of-week breakdown. Friday requires the highest number of drivers, 240 under the base scenario, more than Saturday (210) or Sunday (208). This is counterintuitive given the perception that weekends are busier. The explanation lies in opening hours: Friday operates on a 11.9-hour platform window compared to Saturday's 13.5 hours, so the same volume of demand is compressed into fewer hours, requiring more concurrent drivers. Monday and Tuesday are the lowest-demand days at 197 drivers each.

The key assumptions (Q5) behind this estimation are stated transparently and justified by the data: city efficiency is treated as a fixed constant of 2.147 orders/hr as monthly variance is negligible (± 0.02) across 48 months (A1); opening hours are calculated as data-validated means of 11.90 hours for weekdays (A2) and 13.50 hours for weekends (A3). The fulfillment rate is assumed to remain at its structural ceiling of ~85% based on the Q2 finding that adding couriers does not improve it (A4). The Holt-Winters MAPE of 4.70% is assumed to hold forward for the forecast period, having been validated against a 120-day test set (A5). For planning purposes, a simplifying assumption is made that one driver works one full shift per day (A6), and the model assumes no structural demand shocks in Q1 2026, with the damped trend moderating extreme extrapolation (A7). Finally, a +10% planning buffer is applied as an industry-standard contingency to cover forecast error and operational variables like illness or no-shows (A8).

Additional Analysis · Revenue Impact & SLA Simulation

To quantify the financial cost of unfulfilled demand, two external benchmarks were applied: an Average Order Value of €26.00, sourced from Just Eat Takeaway's 2023 reported figures via businessofapps.com, and a platform commission rate of 18%, implied from JET's 2023 financials (€3.1B revenue divided by €17.0B GMV equals 18.2%, rounded to 18%). These give a revenue per order of €4.68 in commission. Applying this to the lost orders in each year:

Year	Orders Delivered	Lost Orders	Lost Commission
2022	1,427,065	250,796	€1,173,725
2023	1,579,893	279,051	€1,305,959
2024	1,947,896	339,147	€1,587,208
2025	2,041,997	354,660	€1,659,809
Total	6,996,851	1,223,654	€5,726,701

The total four-year lost commission is €5,726,701, approximately **€5.7 million**. In 2025 alone, 354,660 unfulfilled orders represent €1.66 million in lost commission at an average of 974 lost orders per day. Each one percentage point improvement in fulfillment rate would generate approximately €112,472 in additional annual commission based on 2025 potential demand levels. Moving from the current 85% structural ceiling to 88% would unlock around €337,000 per year. The SLA simulation further shows that a 2-minute reduction in handling time would push 25-minute compliance from 57.5% to 90.7%, an improvement of 33.2 percentage points and 484 additional compliant days. There is also a measurable link between SLA performance and customer rating: on days where the 25-minute target was met, the average delivery_rating was 3.621 stars, compared to 3.421 stars on

missed days, a gap of 0.200 stars. Using a conservative industry benchmark of 5% retention uplift per star, this gap translates to approximately 1.0% additional repeat order rate.

| Pre-Analysis Hypotheses: Results

Five hypotheses were stated before any charts were produced or numbers examined, to prevent unconsciously selecting patterns that confirm existing assumptions. Four of the five were disproved, and the counterintuitive results are among the most analytically significant findings in this study.

H1 Weekends have worse SLA compliance due to demand spikes

Result: Disproved

The correlation between is_weekend and SLA compliance is 0.04, essentially zero. Weekend days perform almost identically to weekdays. Day of week is not a meaningful predictor of SLA deviation.

H2 Assign time is the more volatile component of delivery time

Result: Disproved

Handling time shows a gap of 2.19 minutes between met and missed days, while assign time shows only 0.62 minutes. Handling time dominates 78% of total delivery time and is the source of variability, not the courier matching step.

H3 Fulfillment rate has improved as the platform matured from 2022 to 2025

Result: Disproved

Fulfillment rate averaged 85.17% in 2022 and 85.31% in 2025, effectively unchanged. The R-squared between working hours and fulfillment rate is 0.0008. The ceiling is structural and driven by demand, not by operational capacity.

H4 City efficiency (orders per working hour) fluctuates seasonally

Result: Disproved

Monthly average efficiency is 2.147 orders per hour with a standard deviation of +/-0.02 across all 48 months. There is no seasonal pattern. Efficiency is flat regardless of weather, month, or year.

H5 Higher delivery speed leads to better customer ratings

Result: Confirmed

Average delivery rating on 25-minute SLA-met days is 3.621 stars versus 3.421 stars on missed days, a statistically meaningful gap of 0.200 stars. Speed is confirmed as a primary driver of customer satisfaction in this dataset.

| Closing Observations

Across the five business questions, several findings stand out as particularly significant. First, Munich's delivery operation appears healthy against its official SLA but is structurally fragile. A mean delivery time of 24.71 minutes against a 25-minute target leaves almost no margin, and compliance has deteriorated every year since 2022. Second, adding courier hours has not and will not solve the fulfillment problem. The data is unambiguous that demand, not supply, is the binding constraint. Third, a single operational change, reducing restaurant handling time by 2 minutes, would transform Munich's SLA performance more than any other lever identified in this analysis. Fourth, the forecasting evidence points to continued order volume growth into 2026, meaning that without structural changes, the lost order and lost commission figures will continue to rise beyond the €5.7 million already accumulated over four years.